WORKED SOLUTION

STEP II 2002, Question 5

Pure • Logistic Map Iteration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q5

The key idea

Everything hangs on the parabola $y = kx(1-x)$, whose largest value on $[0,1]$ is $\frac{k}{4}$ at $x = \frac{1}{2}$. That alone settles part (i) by induction. Parts (ii) and (iii) ask for fixed points and then points of period two. The trick in (iii) is to write x_1 in terms of b: $x_1 = ka(1-a) = ab$. Then $x_2 = a$ turns into a cubic in b, and that cubic has an obvious root which is exactly the fixed point from (ii).

Solution

Part (i)

We use induction on n. The case $n = 0$ is given. Suppose $0 < x_n < 1$. Then $x_n > 0$ and $1 - x_n > 0$, so $x_n(1-x_n) > 0$. Completing the square,

$$x_n(1-x_n) = \tfrac{1}{4} - \left(x_n - \tfrac{1}{2}\right)^2 \leq \tfrac{1}{4}.$$

Since $0 < k < 4$, multiplying through by k keeps the inequalities, and

$$0 < x_{n+1} = kx_n(1-x_n) \leq \tfrac{k}{4} < 1.$$

So $0 < x_{n+1} < 1$, and by induction $0 < x_n < 1$ for all n. Notice where each hypothesis is used: $k > 0$ for the lower bound and $k < 4$ for the upper one.

Part (ii)

A constant sequence needs only $x_1 = x_0$, that is $a = ka(1-a)$. Since $a \neq 0$ we may divide by a to get $1 = k(1-a)$, and since $a \neq 1$ we may divide by $1-a$:

$$k = \frac{1}{1-a}.$$

Conversely, with this k we have $x_1 = a$, and so every later term is a too.

Part (iii)

With $b = k(1-a)$ we have

$$x_1 = ka(1-a) = ab.$$

If $x_2 = a$ the sequence repeats with period two, so the whole condition is just $x_2 = a$:

$$k\,(ab)(1-ab) = a.$$

Divide by $a \neq 0$ to get $kb(1-ab) = 1$. Since $a \neq 1$ we can write $k = \dfrac{b}{1-a}$, so

$$\frac{b^2(1-ab)}{1-a} = 1 \quad \Longrightarrow \quad b^2 - ab^3 = 1-a \quad \Longrightarrow \quad ab^3 - b^2 + (1-a) = 0,$$

which is the required result.

Ask yourself Which root of the cubic can we spot, and what does it mean?

Putting $b = 1$ gives $a - 1 + 1 - a = 0$, so $b = 1$ is a root. But $b = 1$ means $k(1-a) = 1$, which is the fixed point of part (ii): then $x_1 = ab = a$. So the extra condition $x_1 \neq a$, which says $ab \neq a$, is exactly $b \neq 1$. Dividing out the factor $b - 1$:

$$ab^3 - b^2 + (1-a) = (b-1)(ab^2 + (a-1)b + (a-1)),$$

which you can confirm by expanding. So b must be a root of

$$ab^2 - (1-a)b - (1-a) = 0$$

other than 1. Since $a \neq 0$, this really is a quadratic, with

$$b = \frac{(1-a) \pm \sqrt{(1-a)^2 + 4a(1-a)}}{2a} = \frac{(1-a) \pm \sqrt{(1-a)(1+3a)}}{2a}.$$

For real roots we need $(1-a)(1+3a) \geq 0$, which means $-\frac{1}{3} \leq a \leq 1$. With $a \neq 0, 1$ this leaves $-\frac{1}{3} \leq a < 1$, $a \neq 0$. For any other value of a no real k works.

In that range $1 - a > 0$, so dividing by $1 - a$ gives $k = b/(1-a)$ in a neat form, using $\sqrt{(1-a)(1+3a)}/(1-a) = \sqrt{(1+3a)/(1-a)}$:

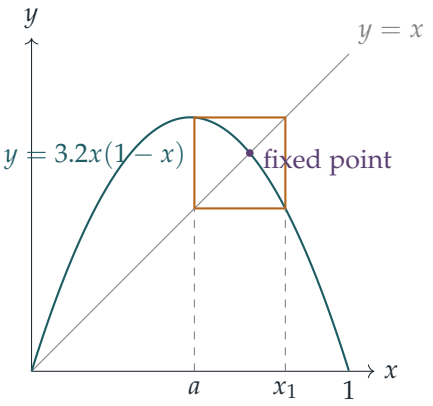
$$\boxed{k = \frac{1}{2a}\left(1 \pm \sqrt{\frac{1+3a}{1-a}}\right), \qquad -\tfrac{1}{3} \leq a < 1,\ a \neq 0.}$$

One last thing: we must make sure neither value is the excluded root $b = 1$. The quadratic has $b = 1$ as a root only when $a + (a-1) + (a-1) = 0$, i.e. $a = \frac{2}{3}$. Then $\sqrt{(1+3a)/(1-a)} = 3$ and the two values are $k = 3$ (which is $b = 1$, the fixed point, so it is excluded) and $k = -\frac{3}{2}$. So:

- for $a = \frac{2}{3}$ the only possibility is $k = -\frac{3}{2}$;
- for $a = -\frac{1}{3}$ the two signs coincide and $k = -\frac{3}{2}$;

- for every other allowed a both signs give a genuine 2-cycle.

The picture below shows one such cycle, for $k = 3.2$. The orbit bounces between $a \approx 0.513$ and $x_1 \approx 0.800$, tracing a square between the parabola and the line $y = x$.



Checks

- Take $a = 0.7$. The formula gives $k \approx 3.0104$ or $k \approx -1.5818$. With $k \approx 3.0104$ we get $x_1 \approx 0.6322$ and then $x_2 \approx 0.7000$, so the orbit really does have period two.
- At $a = \frac{2}{3}$, $k = 3$ gives $x_1 = 3 \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{3}$, the fixed point, as predicted. With $k = -\frac{3}{2}$ we get $x_1 = -\frac{1}{3}$ and $x_2 = -\frac{3}{2} \cdot (-\frac{1}{3}) \cdot \frac{4}{3} = \frac{2}{3}$, a genuine 2-cycle.
- Numerically, $x_2 = a$ held for thousands of random $a \in [-\frac{1}{3}, 1)$ with both signs. For a outside that interval, a scan over k found no solution except the fixed point $k = 1/(1-a)$.

Common slips

- Forgetting that the cubic must contain the fixed-point solution. Spotting the root $b = 1$ is what makes the cubic easy.
- Leaving out the reality condition $-\frac{1}{3} \leq a < 1$, or not noticing that at $a = \frac{2}{3}$ one of the two values of k is the fixed point and must be thrown away.
- In (i), proving only $x_{n+1} \leq \frac{1}{4}k$ without noting this is strictly less than 1 because $k < 4$.

Beyond STEP

This is the logistic map, the classic example of a simple rule with very rich behaviour. For k in the usual range $0 < k \leq 4$ the 2-cycle you found exists exactly when $k > 3$: at $k = 3$ the fixed point loses stability and splits into two points. This is the first

period-doubling bifurcation. Further doublings follow, at k-values that bunch together geometrically (at a rate set by Feigenbaum's constant), until the map becomes chaotic. Strogatz's *Nonlinear Dynamics and Chaos* tells the whole story at first-year level.

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